Module 4: Data Security

Module Objectives

At the end of this course, you will be able to:

1. Understand Salesforce data security importance.

2. Master data security features in Salesforce.

3. Control access to Salesforce objects and records.

4. Manage role hierarchies for efficient access.

5. Define sharing rules for extended access.

6. Apply skills through practical exercises.

Introduction to Data Security

Data security is a critical aspect of information management, ensuring the confidentiality, integrity, and availability of data.

In Salesforce, data security measures are implemented to protect sensitive information from unauthorized access, misuse, or breaches.

Effective data security practices are essential for maintaining trust, compliance with regulations, and safeguarding valuable organizational assets.

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User License: This is assigned to any user of the org. It depends on the user
requirement, which license to opt for.
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Data Security: To avoid any unauthorized access from any user to the org,
object, field, tab, record etc. We use data security.

We have 4 levels of security:
1. Organization level security: We can implement this in below 3 ways:
--> Password policies:
--> Login Hours --> Users can login to their salesforce org only during the
specific number of hours. 9am to 5pm
--> Login IP Ranges
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Profile is of 2 types:
        --> Standard Profile --> It is created by salesforce and we cannot
edit most of the permissions from a standard profile.
        --> Custom Profile --> to edit the permissions, we clone the standard
profile and we create a custom profile, which we can easily edit.
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Difference between Profile and Permission Set:

Scenario : HR Department

Create a profile for a particular group of users. We create one single profile
for whole HR department. We assign that profile to all 5 people below.
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In that profile--> we have Read, edit, create and delete permission.

HR Senior Manager --> Read, create, edit, delete
HR Manager--> Read, create, edit, delete
HR Analyst--> Read, create, edit, delete
HR Analyst--> Read, create, edit, delete
HR Analyst--> Read, create, edit, delete
```

Only want to give read permission to analyst, to give different permission we will create diff profile for permission, but its not good to create new profile to give single permission.

We say give minimum level of access to permission in profile and assign it to all of them,

To give addition permission, we can create **permission set**.

Profile = min. level of permission

Permission set = additional permission

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Difference between Profile and Permission Set: Profile is used to give minimum
level of access to the users. whereas permission set is used to give
additional permissions(additional to profile)
```

Security Data Model

Security data model in Salesforce is structured into four levels, each level control what user can see and do with the data:

Level	Description
Org Level	It determines which users have login access based on the IP ranges and login hours, also it has the password policies and session settings to configure.
Object-Level	Controls access to specific types of data objects, such as Leads, Contacts, Opportunities, and Custom Objects. Administrators can define object CRUD (Create, Read, Update, Delete) permissions.
Record-Level	Determines which individual records users can access within an object.
Field-Level	Specifies which fields within a record users can view and edit. Field-level security allows administrators to restrict access to sensitive information by controlling which fields are visible and editable based on user profiles or permission sets.

Manage Object permission by Creating/assigning profiles

- A Salesforce profile is a set of permissions and settings that determine what a user can see and do within the Salesforce platform.
- Each user in Salesforce is assigned a profile, which determines their access and permissions.
- Salesforce provides **Standard Profiles** with predefined settings, and administrators can also create **Custom Profiles** by cloning Standard Profile to tailor access permissions to specific user job functions within their organization.
- Each profile is associated with a license Type. Example : Salesforce, Salesforce Platform, Chatter Free.

Standard and Custom profile

Standard Profiles	Custom Profiles
Predefined by Salesforce.	Created by administrators.
Determines baseline access for Standard users defined by Salesforce.	Addresses specific access requirements for unique user job functions. . Ex: Recruiter..
Configured with common job functions like System Administrator, Standard User, and Marketing User.	Allows finetuning of permissions.
Cannot customize the Permission	Can customize the Permissions

Open-Up Object permission by using Permission set

Permission Set Groups:

- Combine multiple permission sets into one group.
- Simplify assigning and managing user permissions.
- Offer flexibility by allowing different permissions for various job functions.

Muting Permissions:

- Fine-tune access by disabling specific permissions within a group.
- Temporarily mute certain permissions without changing the original sets.
- Provide detailed control over user access.
- Useful for restricting or removing unneeded permissions.

Managing Object permission by using Permission set

Profiles	Permission Sets
Assigned to all users	Assigned to single/multiple users on top of the Profile.
Defines user permissions and access	Extends permissions in addition to profile settings
Less flexible	More flexible for customization
Profiles will have restrictive access	Permission sets will open restrictions for specific users from profiles.
User will have a single Profile	User can have multiple Permission sets.

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4. Record level security: it is used to avoid unauthorized access for the
records or data. This can be done in 4 ways:
--> OWD(Organization Wide Defaults)
--> Role Hierarchy
--> Sharing Rules
--> Manual Sharing
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--> OWD(Organization Wide Defaults): We have 3 basic access levels in OWD:
        --> Private: the records will be visible and editable only to the
record owner.
        --> public read only: the records will be visible to everyone but
editable only by the record owner.
        --> public read write: the records will be visible and editable by
everyone.
```

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        --> Controlled by Parent: this access level will be available in the
child objects only. ()

position ---> Job application
Parent --> position
Child --> job application --> Controlled by parent --> means whatever access
we give to its parent object, same access will be applied to this object as
well.
```

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        --> Public Read/Write/Transfer --> this is only available in lead and case objects. These are standard objects.
```

Setup Audit Trails

Purpose of Audit Trail

The Setup Audit Trail is a feature that enables administrators to track changes made to their organization's setup configuration.

This includes modifications to various settings, such as profiles, permissions, custom objects, flows & record triggered flows and Apex triggers..

The Setup Audit Trail provides a detailed log of these changes, including the date and time of the change, the user who made the change, and the specific components that were modified.

By enabling the Setup Audit Trail, administrators can maintain visibility and control over their Salesforce organization's configuration, ensuring compliance with internal policies and regulatory requirements.

It also helps in troubleshooting issues, understanding the impact of changes, and reverting to previous configurations if necessary.

Setup Audit Trails

Purpose of Audit Trail

Setting	Description
Profiles	Tracks changes to user profiles, including permission sets, login hours, and assigned roles.
Permissions	Records modifications to object permissions, field-level security, and record types.
Custom Objects	Logs changes to custom object definitions, fields, validation rules, and page layouts.
Workflows and Triggers	Tracks alterations to flows & record triggered flows and Apex triggers.
Settings and Setup	Records changes to general settings, such as organization-wide defaults and email templates.

Field History Tracking

Purpose of Field history tracking

Field history tracking in Salesforce is to enable organizations to keep track of changes made to a specific field within their Salesforce records.

This feature allows administrators to monitor and audit modifications to data over time.

It providing a detailed history of who made the changes, when they were made, and what are the previous and new values.

Field history tracking serves several key purposes:

- **Compliance and Governance**
- **Data Integrity**
- **Performance Analysis**
- **Troubleshooting**

Transferring Records

Mass Transfer Records

Transferring records involves the process of reassigning ownership of multiple records from one user to another simultaneously.

This functionality is particularly useful when there is a change in organizational structure, or ownership responsibility within the Salesforce organization.

The transfer of records ensures that the appropriate users have responsibility for managing specific records, maintaining data integrity and workflow continuity.

Instead of transferring records individually, administrators can select a group of records and transfer them in bulk, saving time and ensuring data accuracy.

Transferring Records

Mass Transfer Records

Aspect	Description
Ownership Change	The process of reassigning ownership of records from one user to another.
Mass Transfer	Supports transferring multiple records simultaneously, beneficial for bulk ownership reassignment or data migration.
Criteria-Based Transfer	Enables automated record transfers based on predefined criteria, triggered by flows, or Record trigger flows.
Auditing and Tracking	Salesforce maintains an audit trail of record ownership changes, including the date, time, and user responsible for the transfer.